

Computer Architecture

2. Data Forwarding Pipeline

2.1 Stage hazards

Branch instructions (be, bne, bl, bge, etc.) present a problem for the pipeline because they can change the program counter (PC) and thus the sequence of instructions. If a branch instruction is in the pipeline, the processor must wait until the branch instruction has completed its execution and the PC has been updated before it can fetch the next instruction. This is because the processor does not know which instruction to fetch next until the branch instruction has completed its execution. This is a problem because the processor is designed to fetch instructions sequentially, and a branch instruction can change the sequence of instructions. To solve this problem, processors use a technique called branch prediction. The processor predicts the outcome of the branch instruction and fetches the next instruction accordingly. If the prediction is correct, the processor can continue to execute the instructions without any delay. If the prediction is incorrect, the processor must flush the pipeline and fetch the correct instruction. This is a problem because flushing the pipeline and fetching the correct instruction takes time, which can slow down the execution of the program. To avoid this problem, processors use a technique called branch target caching. The processor caches the target address of the branch instruction and fetches the instruction from the cache. This avoids the need to flush the pipeline and fetch the correct instruction, which can speed up the execution of the program.